f) $\neg(p \to q) \to \neg q \equiv \neg(\neg(p \to q)) \vee \neg q$ (By conditional-disjunction equivalence)

$\equiv (p \to q) \vee \neg q$ (By double-negation Law)

$\equiv (\neg p \vee q) \vee \neg q$ (By conditional-disjunction equivalence)

$\equiv \neg p \vee (q \vee \neg q)$ (1st associative Law)

$\equiv \neg p \vee T$ (1st negation Law) $\equiv T$ (1st Domination Law)

16) Show that each conditional statement in Exercise 12 is a tautology by applying a chain of logical identities as in Example 8 (Do not use truth tables.)

a) $[\neg p \wedge (p \vee q)] \to q \equiv \neg(\neg p \wedge (p \vee q)) \vee q$ (Conditional-disjunction equivalence)

$\equiv (\neg(\neg p) \vee \neg(p \vee q)) \vee q$ (1st De Morgan's Law)

$\equiv (p \vee \neg(p \vee q)) \vee q$ (Double-negation Law)

$\equiv (\neg(p \vee q) \vee p) \vee q$ (1st commutative Law)

$\equiv \neg(p \vee q) \vee (p \vee q)$ (1st Associative Law)

$\equiv (p \vee q) \vee \neg(p \vee q)$ (1st Commutative Law)

$\equiv T$ (1st Negation Law)

b) $[(p \to q) \wedge (q \to r)] \to (p \to r) \equiv \neg[(p \to q) \wedge (q \to r)] \vee (p \to r)$ (Conditional-disjunction equivalence)

$\equiv [\neg(p \to q) \vee \neg(q \to r)] \vee (p \to r)$ (1st De Morgan's Law)

$\equiv [\neg(\neg p \vee q) \vee \neg(\neg q \vee r)] \vee (\neg p \vee r)$ (3 applications of Conditional-disjunction equivalence)

$\equiv [(\neg(\neg p) \wedge \neg q) \vee (\neg(\neg q) \wedge \neg r)] \vee (\neg p \vee r)$ (2 applications of 2nd De Morgan's Law)

$\equiv [(p \wedge \neg q) \vee (q \wedge \neg r)] \vee (\neg p \vee r)$ (2 applns. of Double-negation Law)

$\equiv [(p \wedge \neg q)] \vee [(q \wedge \neg r) \vee (\neg p \vee r)]$ (1st Associative Law)

$\equiv [(p \wedge \neg q)] \vee [(\neg p \vee r) \vee (q \wedge \neg r)]$ (1st Commutative Law)

$\equiv [(p \wedge \neg q)] \vee [\neg p \vee (r \vee (q \wedge \neg r))]$ (1st Associative Law)

$\equiv [(p \wedge \neg q)] \vee [\neg p \vee ((r \vee q) \wedge (r \vee \neg r))]$ (1st Distributive Law)

$\equiv [(p \wedge \neg q)] \vee [\neg p \vee ((r \vee q) \wedge T)]$ (1st Negation Law)

$\equiv [(p \wedge \neg q)] \vee [\neg p \vee (r \vee q)]$ (1st Identity Law)

$\equiv [(p \wedge \neg q) \vee \neg p] \vee (r \vee q)$ (1st Associative Law)